Slip 12

Q1) A What specific digital evidence can be obtained from browser history analysis in digital forensics?

1. Introduction

Browser history analysis is a digital forensic technique used to examine web browser records to identify a user's online activities. Investigators analyze browser history, cache, cookies, downloads, and related records to collect digital evidence during cybercrime investigations.

2. Digital Evidence Obtained from Browser History Analysis

1. Visited Websites: Records of websites and URLs visited by the user. These can help identify websites accessed during an investigation.

2. Date and Time of Visits: Timestamps showing when a website was accessed. These help establish a timeline of online activities.

3. Search History: Records of searches made using search engines. These may provide information about the user's interests or activities relevant to a case.

4. Downloaded Files: Information about downloaded files, including file names, download locations, source URLs, and download times, when recorded by the browser.

5. Cookies: Small data files stored by websites that may contain session identifiers, preferences, or tracking information.

6. Cached Data: Copies of website resources, images, and other content stored temporarily by the browser. These may help reconstruct previously accessed web pages.

7. Login and Session Evidence: Browser records may reveal evidence of website access and active sessions. Passwords are not necessarily stored or recoverable.

8. Bookmarks and Saved Pages: Saved website links and web pages that may indicate websites of interest or provide evidence related to an investigation.

3. Importance in Digital Forensics

- Helps reconstruct a user's online activity timeline.
- May identify access to suspicious or malicious websites.
- Can support investigations involving fraud, cyberstalking, or unauthorized access.
- Helps correlate browsing activity with other digital evidence.

4. Precautions

Investigators should preserve the original evidence, maintain a proper chain of custody, use forensic tools, and verify findings with other evidence. Browser records may be deleted, incomplete, shared between users, or affected by private browsing modes. A browser history entry alone does not prove who performed the activity or establish criminal intent.

Conclusion

Browser history analysis provides valuable evidence such as visited URLs, timestamps, search queries, downloads, cookies, and cached content. These records help investigators reconstruct online activities and support digital forensic investigations.

Q2) B Write a simple Python program to hide a text message inside an image and display the hidden message again.

```
from PIL import Image
import stepic


# Function to hide message
def hide_data():

    # Ask user for original image
    image_name = input("Enter the original image name: ")
```

```
    # Open image
    img = Image.open(image_name)

    # Ask user for secret message
    secret_msg = input("Enter the secret message: ")

    # Convert message into bytes
    data = secret_msg.encode("utf-8")

    # Hide message inside image
    stego_img = stepic.encode(img, data)

    # Ask for output image name
    output_name = input("Enter name for stego image (use .png):
")

    # Save as PNG
    stego_img.save(output_name, format="PNG")

    print("Message successfully hidden!")
    print("Stego image:", output_name)


# Function to retrieve message
def retrieve_data():

    # Ask user for stego image
    image_name = input("Enter the stego image name: ")

    # Open stego image
    stego_img = Image.open(image_name)

    # Decode hidden data
    decoded_data = stepic.decode(stego_img)

    # Convert bytes to normal text
    print("Retrieved Message:", decoded_data.decode("utf-8"))

# Main program
hide_data()
retrieve_data()
```

Or

Q2) B Calculate and encrypt the number $M = 5$ using the RSA algorithm given $p = 3$, $q = 11$, and $e = 7$.

```
p = 3
q = 11
e = 7
M = 5

n = p * q

C = (M ** e) % n

print("Public key:", e, n)
print("Original message:", M)
print("Encrypted message:", C)
```